Customer Behaviour Marketing Analytics

Executive Summary

This analysis identifies different spending behaviors across age groups, income levels and customer involvement patterns. Older consumers (40+) shop in-store more frequently than younger ones, indicating the necessity for in-store promotions aimed at them, whilst younger shoppers may be encouraged through loyalty benefits and experiential marketing. Income does not significantly impact promotion acceptance, indicating that behavioral and psychological aspects such as urgency-based offers may be more successful than income-based targeting. Furthermore, clients who have been inactive for longer prefer to spend more on luxury items when re-engaged, indicating a potential for targeted campaigns with unique premium deals.

The PCA analysis improves consumer segmentation, with PC5 identifying promotion-sensitive clients who engage with many offers but are less likely to purchase in-store. Clustering research highlights certain customer categories, such as luxury-focused purchasers, budget-conscious shoppers, and health-conscious consumers, allowing for more targeted marketing techniques. Businesses could utilise this information to give high-end offers to luxury spenders, low-cost bargains to price-sensitive clients, and wellness-based incentives to health-conscious buyers. Smart Fresh Retail can increase engagement and sales across many consumer categories by connecting promotional activities with customer behavior.

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Introduction

"Marketing analytics is the process of gathering and analysing data to improve the user experience, increase ROI, and build successful strategies. Businesses may use first-party data to better analyse consumer interactions through surveys or behavioral patterns and make modifications to enhance engagement and conversions. It also allows for calculating the financial return of certain marketing efforts, determining which path generates the most sales, and properly deploying resources. Furthermore, analysing past performance enables organisations to identify successful techniques, modify plans, and remove unsuccessful attempts to achieve their objectives more effectively. Overall, marketing analytics provides data-driven decision-making, increasing ROI and marketing performance" (Cote, 2021).

This research paper will conduct the marketing analysis for "Smart Fresh Retail".

Dataset:
https://drive.google.com/drive/folders/1a1Owu8D8H-EzP9VQMwiHKfPJib8zbxI5?usp=drive_link

Data Analysis & Cleaning

Customers born before 1950 may have different spending patterns and be less involved with digital marketing, therefore they were eliminated to ensure an appropriate study. These records were filtered out using R (Figure 1) to enhance marketing insights and consumer segmentation, providing a more accurate depiction of "Smartfresh Retail" target audience.

```
# Eliminating people who born before 1950
retail <- retail[retail$Year_Birth >= 1950, ]
```

Figure 1: Eliminating people who were born before 1950

Created a new column "Age" (Figure 2) for better analysis

```
# Create Age column
data <- data %>%
  mutate(Age = 2015 - Year_Birth)
```

Figure 2: Created Age column

Imputation and Removal:

Imputation and removal are the two primary methods for correcting missing data. Removal refers to deleting missing data pieces to reduce bias. This is useful if there are still enough observations and the missing data is random. The analysis could not be reliable if too much data is eliminated. Imputation, which involves assuming approximate values for missing variables, is effective when just a tiny portion of the data is missing. However, the findings might not be accurate if too much is missing. In addition, using Mean or Median helps minimise variation and ignores relationships among variables (edX, 2023). The dataset has only 23 null values, so going with the Imputation (Figure 3) is the best option.

```
library(dplyr)

# Converting blank values to NA
retail$Annual_Income[retail$Annual_Income == ""] <- NA

# Converting the column to numeric
retail$Annual_Income <- as.numeric(retail$Annual_Income)

# Calculate the mean, excluding NA values
mean_income <- round(mean(retail$Annual_Income, na.rm = TRUE), 2)

# Replace NA values with the mean income
retail$Annual_Income[is.na(retail$Annual_Income)] <- mean_income
```

Figure 3: Code for replacing null values with mean value

Recency:

Recency refers to the period since a customer's previous interaction (agilebrandguide, 2024), such as a purchase or website visit. Customers who engage currently are more likely to respond to marketing, therefore recency is an important statistic for targeting active and interested consumers (Delval, 2021). Because luxury items have longer purchase cycles, some clients return after over 90 days. To keep this analysis relevant, eliminated data where luxury spendings are less than £10 and the last interaction greater than 60 days (Figure 4). This helps to focus on active purchasers while avoiding misleading insights.

```
#Deleting the where Last_Interaction > 60 & Spend_LuxuryGoods < 10
retail <- subset(retail, !(Last_Interaction > 60 & Spend_LuxuryGoods < 10))
```

Figure 4: Code for deleting Last_Interaction>60 & Spend_LuxuryGoods<10

Analysis of Central Tendency and Variance

In this analysis, these are key attributes to focus on:

- **Annual Income:**
  When customers make more money, they buy more goods and upgrade products. If businesses sell more, they can recruit more employees. When salaries decline, consumers spend less, choose a cheaper alternative, and buy second-hand goods. Businesses sell less and may cut production or jobs (BBC, 2025).
- **Spend Luxury Goods:**
  Luxury brands are growing fast and leading in sustainability and digital transformation. Despite global challenges like rising costs and environmental concerns, the market rebounded strongly in 2022. To stay relevant, brands are shifting strategies, giving consumers more influence over products and marketing (Brittany, 2022).
- **Last Interaction**:
  The recency (Last interaction (agilebrandguide, 2024)), effect influences product perception and success because people prioritize their most recent experiences. This cognitive bias impacts preferences and evaluations, therefore, recent relationships are critical for influencing customer decisions (Braithwaite, 2025).

- **Purchases in Store:**
  In-store sales remain important in retail, allowing customers to explore, test items, and engage with employees. To increase sales, businesses should provide a personalised, engaging experience that includes excellent service, product availability, and unique or exclusive deals (Matthews, 2023).

	Mean	Median	Variance	Std_Dev
Annual_Income	53064.597232	53083	6.453245e+08	25403.237862
Spend_LuxuryGoods	48.401884	28	2.850745e+03	53.392370
Last_Interaction	45.680795	45	7.974645e+02	28.239415
Purchases_Store	5.965463	5	1.050875e+01	3.241721

Figure 5: Output of Central tendency and variance for selected Variables

The analysis below is based on both Figure 5 & Figure 6

The annual income histogram is right-skewed, with a range, demonstrating income disparities among consumers. Spending on luxury goods is likewise skewed, with a median (28) less than the mean (48.4), indicating that a few high spenders influence the average. Last interaction has a near uniform distribution, with a mean (45.68) close to the median (45), indicating that the consumer's involvement varies over time. Purchasing in store has a mean of 5.96 and a low variation, indicating that most consumers make the same number of store visits. The density curves demonstrate skewness in income and expenditure patterns, which are critical for segmenting high-value clients and optimising marketing strategies.

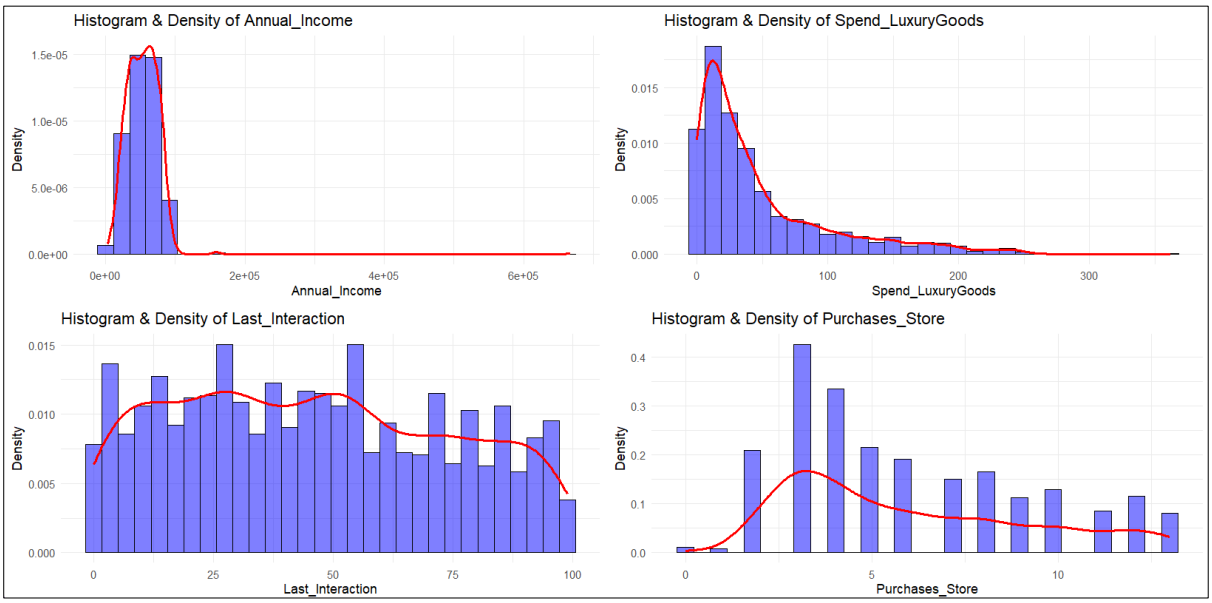

Figure 6: Histograms for selected variables

T-test for relation between variables

I. Do younger vs older customers spend differently in-store?

Created an estimated age group "Young" and "Old" (Age < 40)

```
        Welch Two Sample t-test

data:  Purchases_Store by Age_Group
t = 3.4566, df = 1185.4, p-value = 0.0005663
alternative hypothesis: true difference in means between group Old and group Young is not equal
to 0
95 percent confidence interval:
 0.2382022 0.8635517
sample estimates:
  mean in group Old mean in group Young
           6.142747            5.591870
```

Figure 7: Output of the relation between the Age group and In-Store spendings

The t-test results (Figure 7) reveal a statistically significant difference in store purchases between young and older consumers (p-value = 0.0005663). Older consumers make an average of 6.14 purchases in store, while younger consumers make 5.59. The 95% confidence interval (0.238 to 0.864) demonstrates that older customers shop more frequently than younger ones. Ignore the null hypothesis since the p-value is less than 0.05, indicating that this difference is unlikely caused by chance.

Older customers should be targeted with in-store benefits as they purchase more regularly in-store. Meanwhile, younger customers may be attracted to visit stores through rewards schemes, discounts or an interesting in-store experience. Understanding this behavior can help improve age-based marketing strategies and increase client engagement.

II. Are customers with higher incomes more likely to accept promotions?

```
        Welch Two Sample t-test

data:  Annual_Income by Accepted_Offer1
t = 1.4372, df = 189.6, p-value = 0.1523
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -1010.592  6436.145
sample estimates:
mean in group 0 mean in group 1
       53281.79        50569.01
```

Figure 8: Output for the relation between Annual income and Accepted Offers

There is no statistically significant difference in the annual income of customers who accepted the promotion compared to those who didn't, according to the t-test (Figure 8). The p-value (0.1523) is greater than 0.05, indicating that the null hypothesis cannot be rejected, which suggests that money has no significant impact on promotion acceptance. Non-acceptors have a mean income of £5328.79, whereas acceptors have a slightly lower average of £50569.0, but the difference is insignificant.

Since income does not significantly impact promotional acceptance, other factors such as product preference, marketing methods, or attractive offer may drive customer decisions. Instead of focusing on high-income customers, this business can consider behavioral and psychological aspects such as urgency-based promotions or personalised discounts.

III. Do customers who recently interacted spend more?

Created a Recent Interaction variable ("Recent" vs "Old") based on whether Last Interaction is above or below the median.

```
        Welch Two Sample t-test

data:  Spend_LuxuryGoods by Recent_Interaction
t = 4.3747, df = 1909, p-value = 1.282e-05
alternative hypothesis: true difference in means between group Old and group Recent is not equal
to 0
95 percent confidence interval:
  5.867306 15.403058
sample estimates:
   mean in group Old mean in group Recent
            53.68887             43.05368
```

Figure 9: Output for the relation between Interaction and Luxury goods

The t-test results (Figure 9) demonstrate a statistically significant difference in luxury expenditure between customers who interacted in the past (p-value = 1.282e-0.5). Deny the null hypothesis since the p-value is much lower than 0.05, demonstrating that the difference in luxury spending is unlikely to be related to chance. Customers with older interaction spend an average of £53.68 on luxury products, whereas those with recent interaction pay £43.05. the confidence interval (5.87 to 15.40) indicates that the gap is consistent.

This finding suggests that customers who have been inactive for longer spend more money on luxury items when they return. Business might profit from tailored re-engagement strategies for inactive clients, such as delivering luxury bonuses or exclusive deals to attract high-value purchases.

Correlation Matrix:

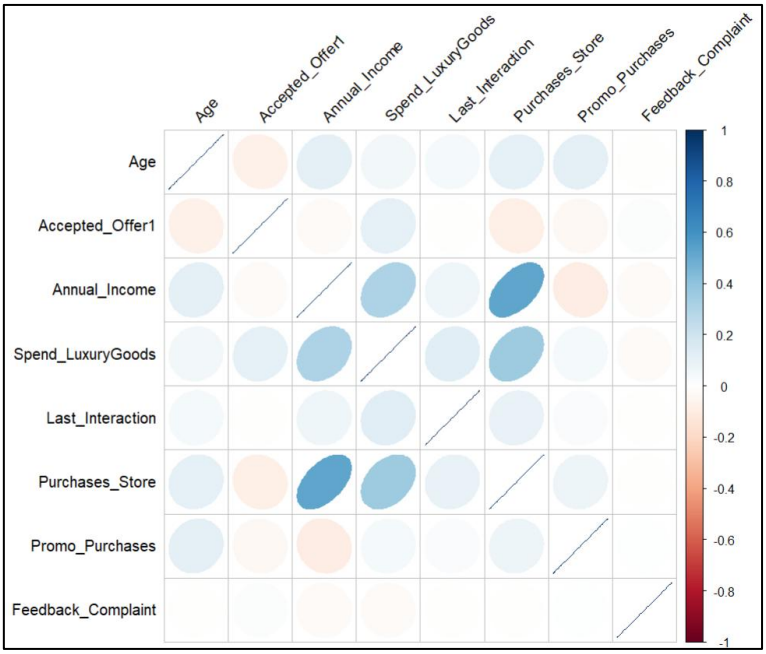

Figure 10: Correlation matrix for selected variables

The correlation matrix (Figure 10) indicates a substantial positive association between Annual income and Spend Luxury Goods, indicating that higher-income clients spend more on luxury products. Purchases Store and Spend Luxury Goods are positively connected, demonstrating

that in-store consumers prefer luxury goods. Promo Purchases correlates with Purchase Store, indicating that promotions. A modest negative association between Age and Accepted Offer1 indicates that younger consumers may respond better to promotions. Feedback Complaint shows poor connections, indicating that complaints do not have a significant influence on spending. Business should target high-income customers for luxury promotions, employ discounts to increase in-store sales, and direct promotional efforts towards younger customers.

Factor Analysis and Interpretation of Dataset Factors

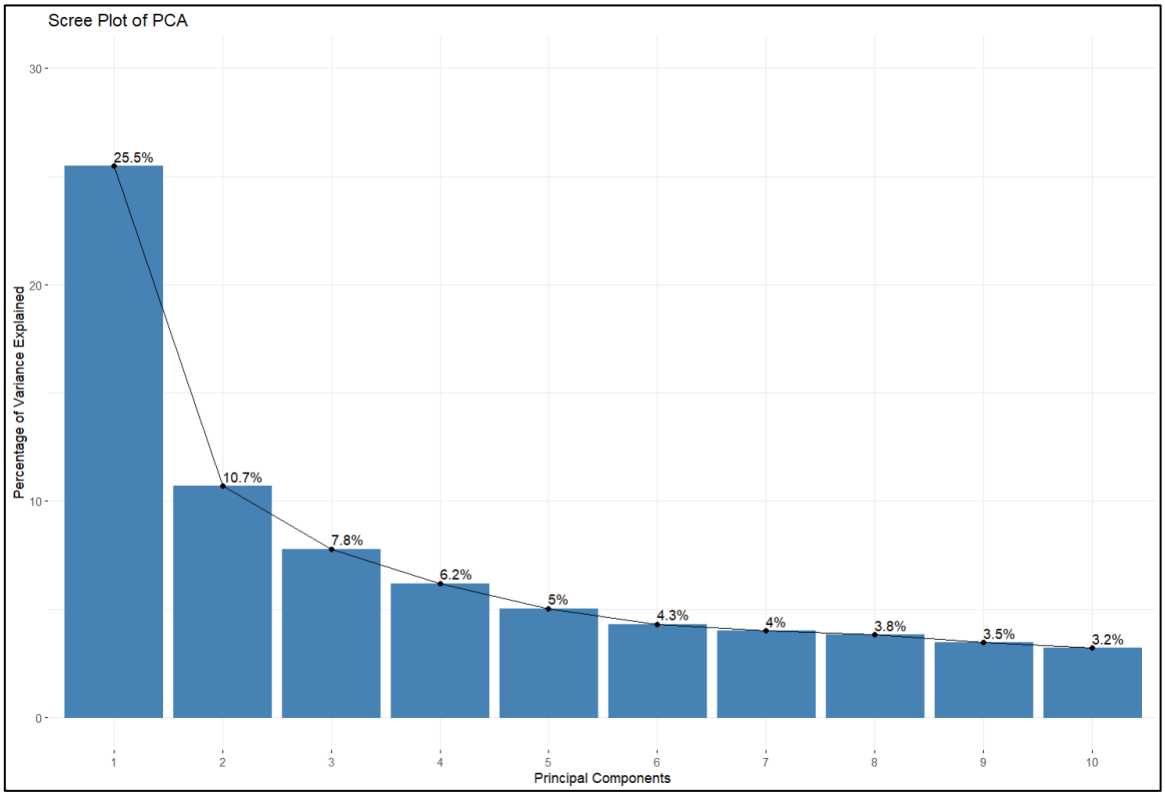

Figure 11: Scree plot of PCA

Scree plot of PCA (Figure 11), which shows that PC1 explains 25.5% of the variance, followed by PC2(10.7%), PC3(7.8%), PC4(6.2%), and PC5(5%). After PC5, the explained variance falls significantly, implying that more components provide less helpful information.

Picking PC5 maintains a balance between variance explained and dimensionality reduction. These five main components capture most of the dataset's variability while filtering out noise from less significant components. This ensures the data's fundamental patterns are preserved while simplifying further analysis.

Now interpret the PCA loadings to discover how each original variable contributes to the selected components. This will help you extract valuable insights from the reduced-dimensional dataset.

	PC1	PC2	PC3	PC4	PC5
Customer_ID	0.00752977	0.003204332	-0.037424399	-0.08027765	-0.035481518
Year_Birth	0.03916575	0.511429482	-0.030563485	0.33533783	-0.230607984
Annual_Income	-0.29084237	-0.057325602	-0.004649529	-0.05032551	-0.085673534
Kidhome	0.25847947	0.111630268	0.058843358	0.15312803	0.069609496
Teenhome	0.05060021	-0.455234804	0.058571457	0.11850682	-0.047308252
Last_Interaction	-0.05303803	-0.064454125	-0.091081726	0.07188138	-0.214052162
Feedback_Complaint	0.01203055	0.007356585	-0.015094190	0.02335386	0.077981468
Spend_Wine	-0.30505467	-0.090150866	0.213273686	0.10381641	-0.124172503
Spend_OrganicFood	-0.26722272	0.075722204	-0.213657422	0.05350099	0.106849110
Spend_Meat	-0.31361719	0.107061224	-0.072412993	-0.03732267	0.046975683
Spend_WellnessProducts	-0.27783974	0.067942400	-0.215464883	0.02008180	0.091561087
Spend_Treats	-0.27054367	0.066663797	-0.174262130	0.05542169	0.054581589
Spend_LuxuryGoods	-0.21704646	-0.057991881	-0.060665509	0.24996553	0.184127528
Purchases_Online	-0.20190105	-0.211451097	0.068103437	0.42229809	-0.027558879
Purchases_Catalog	-0.31957087	-0.005278182	-0.019451415	0.02681596	0.083062653
Purchases_Store	-0.28213117	-0.132284780	-0.064837755	0.15358714	-0.219147113
Visits_OnlineLastMonth	0.25114921	-0.066801065	0.212270169	0.31799007	0.057439837
Promo_Purchases	0.06402804	-0.274631952	0.081818310	0.52282203	-0.004740648
Accepted_Offer1	-0.01031669	0.099515495	0.180298863	0.13636413	0.571433674
Accepted_Offer2	-0.09957303	-0.034605759	0.433511638	-0.07591323	-0.402057397
Accepted_Offer3	-0.19259929	0.153956227	0.342293378	-0.13227390	-0.023918899
Accepted_Offer4	-0.17324401	0.122116789	0.308392461	-0.10416114	0.078218806
Accepted_Offer5	-0.06161301	0.016349324	0.405994844	-0.09644637	-0.165614490
Response_Latest	-0.09512011	0.140788323	0.382001872	0.05305252	0.412842627
Age	-0.03916575	-0.511429482	0.030563485	-0.33533783	0.230607984

Figure 12: PCA loadings values for the dataset

- High absolute values (closer to 1 or -1) → Strong contribution of that variable to the principal component.
- Positive values → Variables positively influence that component.
- Negative values → Variables negatively influence that component.

The PCA loadings (Figure 12) show how different factors influence the five principal components (PCs). PC1 is driven by purchasing behavior, with customers who spend more on meat, organic food, treats, Wine, wellness products, Luxury goods, and store purchases experiencing significant negative loadings.

PC2 demonstrates demographics and offers acceptance, with Year_Birth being the most significant element. Older consumers and those who choose offers 1 and 3 contribute positively, but those with kids and high organic food expenditure contribute adversely.

PC3 focuses on promotional response, with strong positive loadings for customers who take offers 3, 4, and Response_latest. Customers who spend more on organic food, meat, and wellness products had lower scores, indicating a distinct buying behavior.

PC4 refers to Online shopping and promotional activities. Customers who shop online and make promotional purchases do better, but elderly consumers have negative loadings, indicating less online shopping activity.

PC5 is significantly impacted by Promotional behavior, with consumers who accepted many promotional offers (Accepted_offer1 to Accepted_offer5) and replied favorably to promotions (Response_latest) showing high positive loadings. However, Annual_Income and Purchases_store show negative loadings, indicating that high-income consumers and those who have recently shopped in-store are less inclined to participate in promotions.

This analysis suggests that PC5 reflects promotions-sensitive clients, with a substantial correlation between offer acceptance and expenditure on organic food, Luxury items, and promotional purchases. This Business may use this information to tailor promotional marketing campaigns to customers who interact with many offers and are more likely to respond positively to promotions.

Clustering Analysis of Customers' Spending Behavior

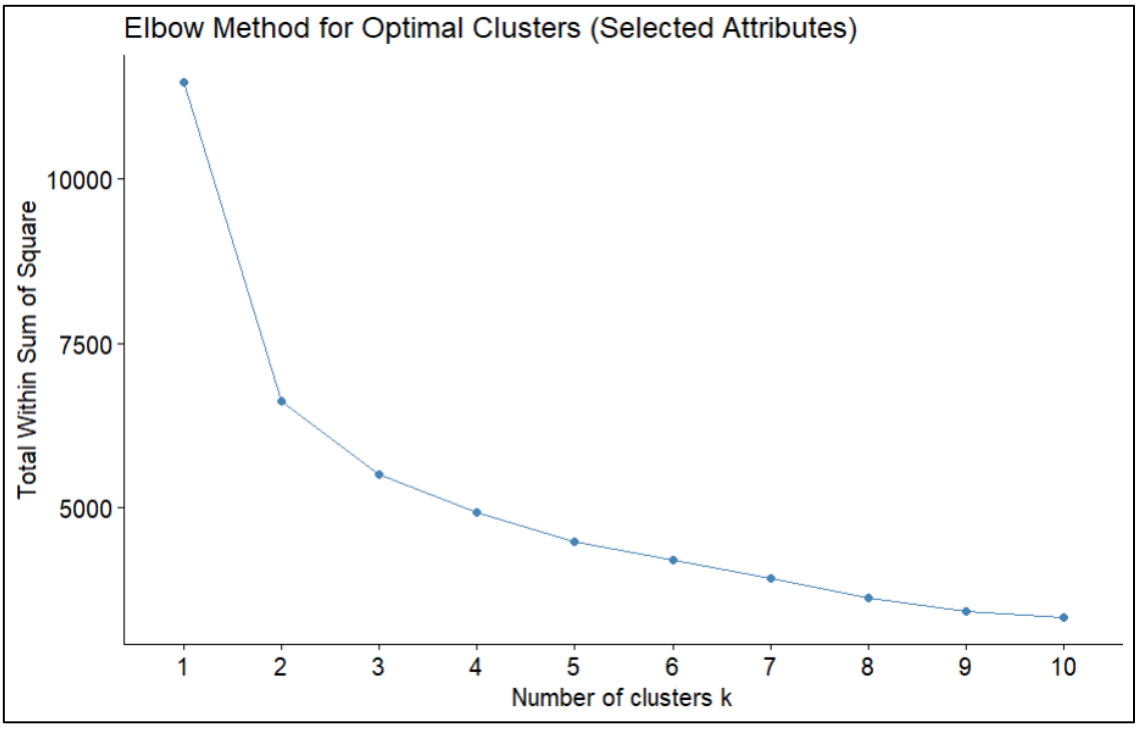

Figure 13: Elbow Method for Optimal Clusters

The elbow method plot (Figure 13) displays the total within-cluster sum of squares (WSS) for various k values and helps determine the best number of clusters. It depicts a large decrease in WSS between 1 and 4, after which the reduction slows. This suggests that four clusters are a suitable number, as adding more clusters would result in a diminishing increase in variance reduction.

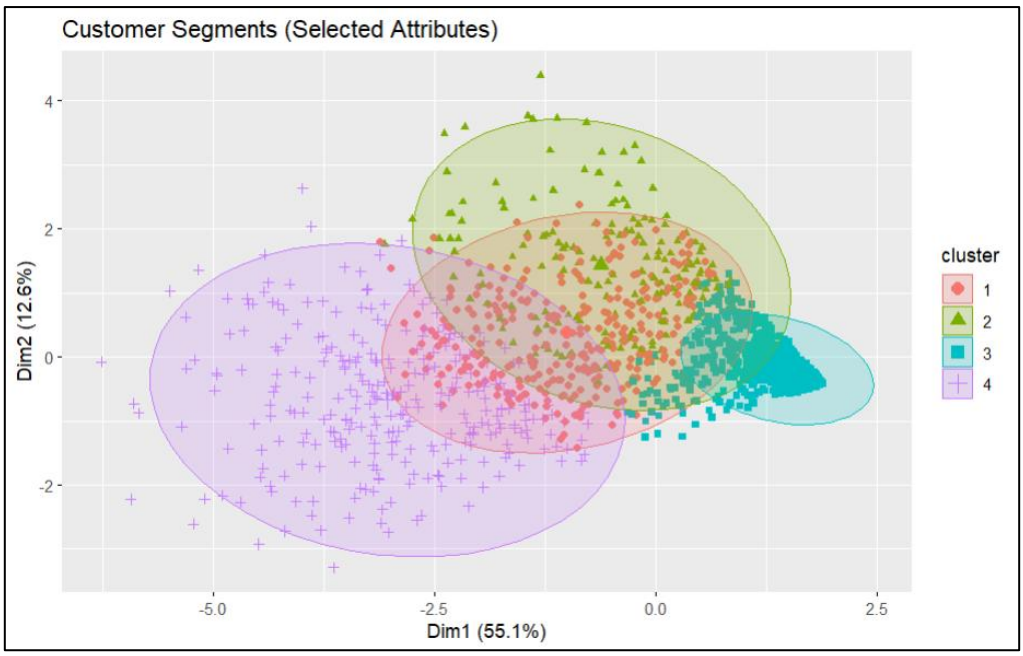

Figure 14: Cluster visualization plot

The cluster visualization plot (Figure 14) displays consumer groupings based on the spending attributes, which are reduced in dimensionality using Principal Component Analysis (PCA). The four clusters are color-coded, and ellipses depict each cluster's distribution. The cluster variation indicates genuine consumer segmentation, with each section having different purchase behaviors. The dispersion and overlap demonstrate that some consumer groups are clearly distinct, others may exhibit slight behavioral similarities.

	Cluster	Spend_Wine	Spend_OrganicFood	Spend_Meat	Spend_WellnessProducts	Spend_Treats	Spend_LuxuryGoods
1	1	714.3988	36.806358	359.11850	52.367052	35.132948	41.79480
2	2	458.4794	25.989691	164.81959	32.932990	25.701031	144.03608
3	3	102.3559	6.898192	38.42912	9.594672	6.956232	19.83635
4	4	514.9188	90.125000	431.40625	131.115625	95.718750	91.38750

Figure 15: Cluster summary table

The cluster summary table (Figure 15) shows the average spending behavior of each cluster. Cluster 1 has the greatest spending on wine and Meat, suggesting high-value food and beverage consumers. Cluster 2 reflects customers with moderate food and treats and the highest spending on luxury goods compared to all clusters. Cluster 3 has the lowest total spendings across all criteria, indicating budget-conscious or low-spending customers. Finally, cluster 4 shows high spending on wellness goods and organic food compared to the other clusters, showing health-conscious customers. These segments might help businesses develop focused marketing efforts, such as premium offerings for high spenders, discount-based promotions for low spenders, and personalised recommendations for health-conscious customers.

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Appendices

Codes:

1.

```
library(ggplot2)

key_variables <- c("Annual_Income", "Spend_LuxuryGoods", "Last_Interaction", "Purchases_Store")

summary_stats <- data.frame(
  Mean = sapply(data[key_variables], mean, na.rm = TRUE),
  Median = sapply(data[key_variables], median, na.rm = TRUE),
  Variance = sapply(data[key_variables], var, na.rm = TRUE),
  Std_Dev = sapply(data[key_variables], sd, na.rm = TRUE)
)

print(summary_stats)


install.packages("ggpubr")

library(ggpubr)

histograms <- lapply(key_variables, function(column) {
  ggplot(data, aes_string(x = column)) +
    geom_histogram(aes(y = ..density..), bins = 30, fill = "blue", alpha = 0.5, color = "black") +
    geom_density(color = "red", size = 1) +  # Density curve
    ggtitle(paste("Histogram & Density of", column)) +
    xlab(column) +
    ylab("Density") +
    theme_minimal()
})

ggarrange(plotlist = histograms, ncol = 2, nrow = 2)
```

2.

I

```
data <- data %>%
  mutate(Age_Group = ifelse(Age < 40, "Young", "Old"))

t_test1 <- t.test(Purchases_Store ~ Age_Group, data = data)
print(t_test1)
```

II

```
t_test4 <- t.test(Annual_Income ~ Accepted_Offer1, data = data)
print(t_test4)
```

III

```
median_interaction <- median(data$Last_Interaction, na.rm = TRUE)

data <- data %>%
  mutate(Recent_Interaction = ifelse(Last_Interaction < median_interaction, "Recent", "Old"))

t_test8 <- t.test(Spend_LuxuryGoods ~ Recent_Interaction, data = data)
print(t_test8)
```

3.

```
num_data <- data1[, sapply(data1, is.numeric)]

num_data <- na.omit(num_data)

if(any(is.na(num_data))) {
  stop("Error: Dataset still contains missing values. Please clean the data.")
}
scaled_data <- scale(num_data)

pca_result <- prcomp(scaled_data, center = TRUE, scale. = TRUE)

if(is.null(pca_result)) {
  stop("Error: PCA computation failed. Ensure data is correctly formatted.")
}

summary(pca_result)

eigenvalues <- pca_result$sdev^2

fviz_eig(pca_result, addlabels = TRUE, ylim = c(0, 30),
         title = "Scree Plot of PCA",
         xlab = "Principal Components", ylab = "Percentage of Variance Explained")

if (!is.null(pca_result)) {
  fviz_pca_biplot(pca_result, repel = TRUE, title = "PCA Biplot")
} else {
  stop("Error: PCA result is NULL. Cannot generate biplot.")
}

cumulative_variance <- cumsum(eigenvalues) / sum(eigenvalues)
print(cumulative_variance)

print(pca_result$rotation)

pca_scores <- as.data.frame(pca_result$x)

data1_pca <- cbind(data1, pca_scores)
```

4.

```
selected_data <- data1[, c("Spend_Wine", "Spend_OrganicFood", "Spend_Meat",
                           "Spend_WellnessProducts", "Spend_Treats", "Spend_LuxuryGoods")]

selected_data <- na.omit(selected_data)

scaled_data <- scale(selected_data)

fviz_nbclust(scaled_data, kmeans, method = "wss") +
  ggtitle("Elbow Method for Optimal Clusters (Selected Attributes)")

set.seed(123)  # For reproducibility
k_optimal <- 4  # Choose based on elbow plot (adjust if necessary)
kmeans_result <- kmeans(scaled_data, centers = k_optimal, nstart = 25)

fviz_cluster(kmeans_result, data = scaled_data, geom = "point",
             ellipse.type = "norm", main = "Customer Segments (Selected Attributes)")

data1$Cluster <- as.factor(kmeans_result$cluster)

cluster_summary <- aggregate(selected_data, by = list(Cluster = data1$Cluster), mean)
print(cluster_summary)
```